

Acoustics: Analogue Filtering

VORTEX NTNU

June 2024

1 Introduction

The pinger sends out pulses with frequencies below 25kHz. We wish to take in signals below half the sampling frequency of the ADC: $f_c = 50\text{kHz}$ and start attenuating everything above this. For the 2.2 PCB version we aimed for a high-order Low Pass Filter with higher fall rate. In addition to the frequency filtering, several other stages were added to remove noise and improve the system.

2 Hydrophones

The hydrophone sensors we are using are **BII-7014FG**. Check the Hydrophone data sheet here: [\[1\]](#) or an alternative at Vortex wiki. Using the Differential Output of the Hydrophones we have

1	Common
2	Signal+
3	VDC
4	Signal-

Where VDC and Common are the power (9V DC) and grounding to the sensors respectively. The fact we have both Signal+ and its exact "inverted" Signal- gives us an advantage with common mode noise rejection (more on that later).

The signal from the Hydrophones is very weak (33mV) and will need additional amplification. The Hydrophone current is also very small (10mA) which is good noise-wise and bad power-wise.

3 Instrumental Amplifier

3.1 Amplifiers

An Operational Amplifier (Opamp) is an IC that amplifies the voltage input or the difference between two inputs: non-inverted A and inverted B. The Opamp has its own supply voltage (V+ and V-) allowing us to amplify even the weakest signals! The IC we use is called the **RC4558** and contains 2 opamps in 1 chip [\[3\]](#).

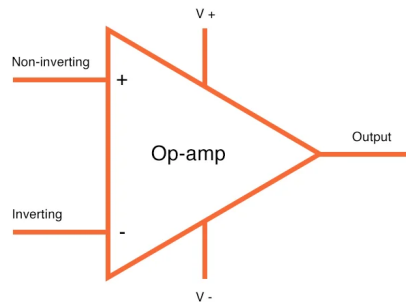


Figure 1: Circuit symbol for an Operational Amplifier

3.2 Common Mode Rejection

Consider inputs: signal A to non-inverting, and a signal B to inverting input of the Opamp. Now if B is already inverted copy of A, the output is going to be $A - (-A) = 2A$. Now lets say both line A and B has some electronic noise. Noise is a random entity which means noise in B will differ from noise in A. That is why the amplification will double the good signal and randomly cancel out the noise. This is not enough to remove all the noise, but its an almost effortless trick.

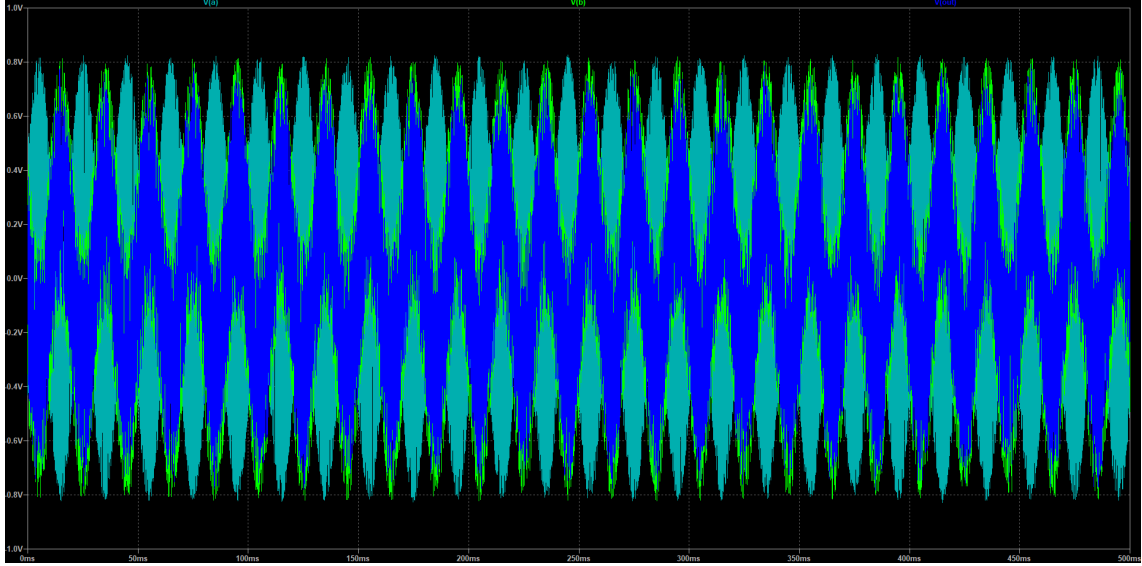


Figure 2: Simulated Common Noise rejection in LTspice. Here b is the inverted version of a. We add White Noise to both signals. Same strength, but noise A has more random values per time interval. The Inamp has 1/2 amplification. Notice the difference between b (lime) and the filtered output (blue)!

Before frequency filtering, we insert an Instrumental Amplifier (Inamp). This setup uses 3 Opamps: the first two for each of the input signals (Signal+ and Signal-) and the last one amplifies the outputs of the first two. For more details read here: [2]. The main takes are the following:

1. The total gain of the Inamp can be changed by simply adjusting R_G (solving the issue with small signal voltage).
2. Common Mode rejection allows us to amplify the the difference between Signal+ and Signal-. The actual acoustic signal gets doubled, while the electronic noise will somewhat cancel out.
3. Another bonus of Inamps is the reducing of loading effects, making our system much more stable and sturdy when it comes to external perturbations.

3 Opamps for each of the five Hydrophones takes too much space, that is why we opted for using a pre-built chip for the Inamp: **INA126**, read the documentation here: [4].

4 Frequency Filtering

Up to this point, the filtering was frequency *independent*, mostly targeting electronic noise. Now we implement a stage that preserves certain ranges of frequencies. The sampling rate of the ADC (digitalization) is 100kHz, so the Cutoff Frequency f_c can not be higher than 50kHz! The pinger produces signals around 23kHz so choosing $f_c = 50\text{kHz}$ is not an issue. The value could be lower, but this leaves us with a safety margin in case for example a new pinger model has higher frequency.

4.1 Low Pass Filters

I link an in-depth guide to design of all kinds of analogue filters here: [5]. We are interested in mainly Low Pass Filters since we want to attenuate everything after 50kHz. As mentioned in the guide, the simplest structure would be passive filters using only resistors and capacitors. This setup would be exposed to power losses, and is therefore unacceptable for our already low-power signal. Active filtering utilises already discussed opamps and is much better suited for acoustics.

We want the fall rate (how fast the gain decreases as the frequency grows) to be as big as possible in the Stop Band ($f > f_c$) while maintaining an even Pass Band ($f < f_c$). Having an even Pass band is very important so that some of the good frequencies don't appear stronger than others. Having both even pass band and high fall rate is the equivalent of sitting on two chairs with one butt! We must look for a compromise. We chose to use a Chebyshev-type filter which boasts the sharpest transition between pass and stop bands and fastest fall rate, but pays with so-called Ripples in the pass band.

To keep the ripples small we have to increase the order of the filter. We use a 4th order filter with a Sallen-Key topology. Sallen-Key topology means we combine several stages in series: 4th order = 2nd + 2nd. You can easily see the order of a stage by the number of capacitors. I should probably mention both inductors and capacitors are viable, but we rather avoid using inductor coils, as they take up much more space, have smaller variety of values and come with larger tolerances.

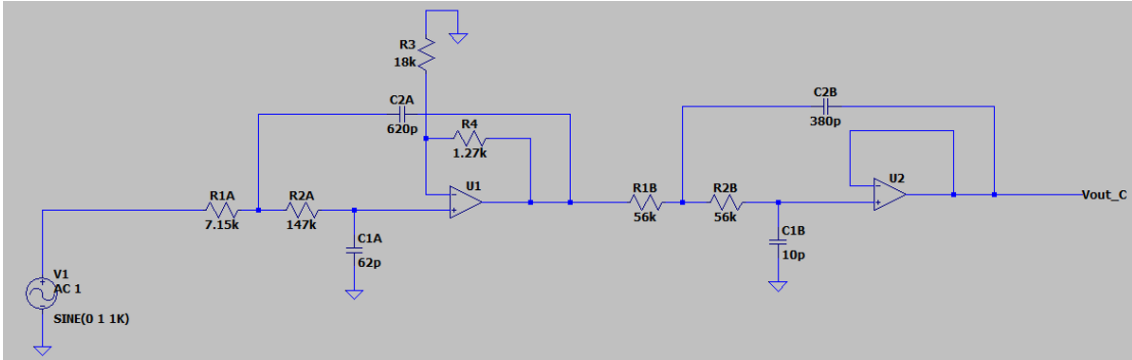


Figure 3: The Circuit of a 4th order, Sallen-Key topology, Chebyshev, LPF. Note my notation for the components. R1A1 encodes to first of 2 resistors that go to non-inverted input of Opamp A of the first of 5 hydrophone-filter sub-systems.

Topologies like this might seem intense, but they base themselves on the simplest RC filters. If its a Low pass filter the output is always "over" the capacitors. You will therefore have capacitors in the feedback loop(s) and resistors along the non-inverted line. If you for some reason wanted to flip this to a HPF just change places between resistors with capacitors. Furthermore if you decided to add more stages 5th 6th... that would mean adding more stages in series. Since our **RC4556** chips contain 2 opamps, each hydrophone requires 1 chip for the inamp and 1 chip for the opamp. A 5th order filter would require an additional opamp and a 3rd order would mean 1 free opamp slot (you end up having 'inelegant' cross wiring between your sub systems).

4.2 Calculating Resistor and Capacitor Values

The guide I have attached gives an explanation how one would go about calculating each and every R and C value for each sub stage of your filter. You find the total transfer function for a nth order Chebyshev filter:

$$H(s) = \prod_{i=1}^{n/2} \frac{1}{(s/a_i\omega_c)^2 + (s/a_i\omega_c)/b_i + 1}$$

Here a_i and b_i are tabulated coefficients you look up for some specific ripple value E [dB] ($1\text{dB} = 20\log_{10}(V_o/V_i)$). Also note the angular cutoff frequency $\omega_c = 2\pi f_c$.

$H(s)$ is the Laplace transform of $h(t)$ that links input $X(s), x(t)$ and $Y(s), y(t)$ together: $Y(s) = X(s)H(s)$ in s-domain and $y(t) = x(t) * h(t)$ in time domain (* denotes convolution). Substituting $s \rightarrow j\omega$ brings us to the frequency domain where we can directly see the attenuation for each frequency. To get all the R and C values you would need to analyse a_i and b_i separately as shown in the guide.

This is an absolute nightmare of a procedure, keep in mind you will have to experiment with different values before you reach a set of values that you can later actually find in stock! Therefore I attach a link to a website that calculates optimal values given a cutoff frequency, ripple and order: [6].

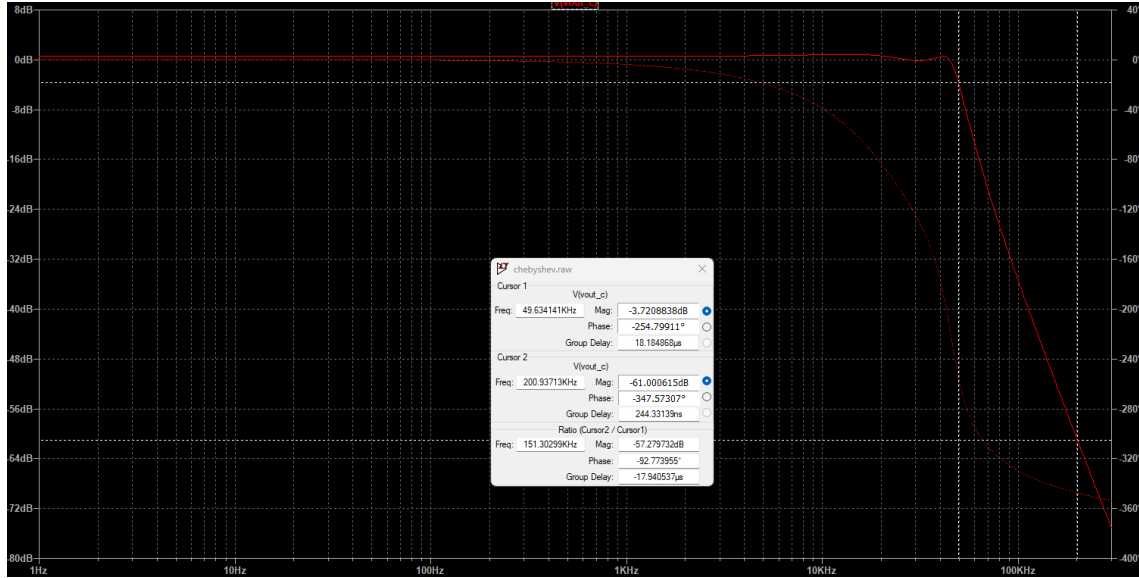


Figure 4: Bode plot for our 4th order Chebyshev filter.

The Bode plot: $20\log_{10}|H(j\omega)|^2$ [dB/Hz] belongs to the circuit in figure 3. This filter was designed to have a cutoff frequency of 50kHz and a pass band ripple of 0.56dB (see the ripple right at the end of the pass band). One of the cursors indicates 50kHz and the corresponding -3dB (cut off gain). Notice how at 200kHz we have 60dB attenuation aka ratio of 10^{-3} to the pass band. This means signals with higher frequency than 200kHz is more than a thousand times weaker than the desired signals!

5 Other Noise Considerations

5.1 Power Supplies, Decoupling and Grounding

Since acoustics is a very sensitive subject slight perturbations can end up blowing up. First of all the power supply. To make sure the 9V VDC is in fact DC we add Pi filters. This removes the ripple present in the supplied voltage. Furthermore we provide two different power supplies for the hydrophones and the filter ICs. We implement decoupling capacitors to separate different parts of the circuit from each other. Decoupling capacitors shunt noise from other circuit elements. When working with high frequency signals one uses ground planes to reduce ground loops when designing the PCB itself. Different ground planes for the analogue and digital part of the circuit is very important.

5.2 Stray Impedance

After deciding on the resistor and capacitor values first of all make sure to check the tolerance of the purchasable items. How does the frequency response change if some of the values are slightly

different. The current design has tolerance accounted for. Thermal noise can become a problem when working with large bandwidth and large resistor values.

$$V_{Th} = \sqrt{4RBWk_BT}$$

Where R is the resistance, BW is the bandwidth of the pass band, T is the expected temperature [K]. For our values the thermal noise lies around $10\mu\text{V}$. Parasitic Inductance is another issue. In the EL-house there are several cables transporting very large currents. These should be kept on a good distance away. Final thing to consider is parasitic capacitance. Our design features some very small value capacitors: 10pF. The design has a margin of 2pF, and it further extendable by reducing R1B and R2B resistors. This margin is larger than what we expect/measured for parasitic capacitance.

References

- [1] Hydrophone data sheet (alternative available on Vortex wiki) <https://www.benthowave.com/products/Specs/BII7071-2-3-4FGPGdatasheet.pdf>
- [2] Instrumental Amplifiers <https://www.elprocus.com/what-is-an-instrumentation-amplifier-circuit-diagram-advantages-and-applications/>
- [3] RC4558 data sheet https://www.ti.com/lit/ds/symlink/rc4558.pdf?ts=1696497051105&ref_url=https%253A%252F%252Fwww.ti.com%252Fproduct%252FRC4558%253Futm_source%253Dsupplyframe%2526utm_medium%253DSEP%2526utm_campaign%253Dnot_alldatasheet%2526DCM%253Dyes
- [4] INA126 data sheet https://www.ti.com/lit/ds/symlink/ina126.pdf?ts=1699340922580&ref_url=https%253A%252F%252Fwww.ti.com%252Famplifier-circuit%252Finstrumentation%252Fproducts.html
- [5] Texas Instruments: Filter design in-depth guide <https://www2.seas.gwu.edu/~ece121/Spring-11/filterdesign.pdf>
- [6] Filter design online tool <https://tools.analog.com/en/filterwizard/>